

Capstone Project Final Report

Predicting High-Value Customers to Prioritize Marketing Spend

Data Science Career Track — Capstone Three

Dataset: Customer Shopping Trends Dataset (Kaggle, ~3,900 customers)

Prepared for: BigMart — Head of Marketing & CRM/Data Analytics Manager

This report summarizes the problem, approach, key findings, model performance, and recommendations resulting from a customer-value segmentation analysis.

1. Problem Statement

Business Question: How can a retail brand (BigMart) increase marketing ROI by predicting high-value customers and personalizing its investment strategy for the upcoming fiscal year?

Context: BigMart currently allocates its marketing budget uniformly across its entire customer base, without differentiating between high, medium, and low-value segments. Features such as purchase amount, purchase frequency, subscription status, and seasonal behaviour suggest significant variation in customer value. A data-driven segmentation model is needed to optimize spend and maximize profitability.

Constraint: The dataset does not include a Customer Lifetime Value (CLV) label, so a proxy metric was required to define 'high-value' segments. (Section 2 explains how this proxy was constructed.)

Business / User Benefits: The resulting model gives the marketing team insight into customer purchasing patterns in order to: predict each customer's value tier (High / Medium / Low), identify the top contributing factors behind that tier (e.g. "subscribed + frequent buyer"), and recommend an action (retention offer, conversion nudge, or leave alone).

2. Approach

2.1 Data Wrangling

The raw dataset (3,900 rows, 19 columns) was checked for missing values, duplicate records, inconsistent categorical labels, and out-of-range / outlier values. The dataset was found to be already clean: zero missing values, zero duplicate rows or customer IDs, consistently formatted categories, and no logically invalid values or IQR-based outliers. No rows were removed.

Because the dataset lacks a CLV label, a proxy **Customer Value Score** was engineered as a weighted combination of the three features most directly tied to customer value:

- Purchase Amount (USD) — weight 0.45
- Previous Purchases — weight 0.35
- Subscription Status (Yes = 1 / No = 0) — weight 0.20

The continuous score was then split into three balanced groups (terciles) using `pd.qcut`, producing the target variable **customer_value_tier** (Low / Medium / High), with roughly 1,300 customers per tier.

2.2 Exploratory Data Analysis (EDA)

EDA examined distributions of key numeric features (Age, Purchase Amount, Review Rating, Previous Purchases), the balance of the three value tiers, and how spend, purchase history, subscription status, season, category, and demographics relate to customer value. A correlation heatmap and grouped comparisons across tiers were used to identify the strongest signals.

2.3 Preprocessing

Before modeling, the following preprocessing steps were applied: dropped the Customer ID, high-cardinality columns (Item Purchased, Location, Color), and the intermediate columns used only to construct the target (to avoid leakage); one-hot encoded the remaining categorical features

(pd.get_dummies, drop_first=True); standardized numeric features (Age, Purchase Amount, Review Rating, Previous Purchases) with StandardScaler; and split the data into an 80% training set and a 20% test set, stratified by tier (3,120 / 780 customers).

2.4 Modeling

Customer value tier is a categorical (3-class, ordinal) target, so this is a **classification** problem — there is no time-series component, so a stratified random train/test split is appropriate. Three classification models of increasing complexity were trained on identical data and compared using the same metrics: Logistic Regression (simple linear baseline), a Decision Tree (max depth 5, easy to interpret), and a Random Forest (100 trees). Models were compared on Accuracy, Macro F1-score, and Mean Absolute Error on the tier codes (Low=0, Medium=1, High=2) — the latter captures how far off a wrong prediction typically is, which matters because the tiers are ordered.

3. Key Findings

Finding 1 — Three features dominate customer value. Purchase Amount, Previous Purchases, and Subscription Status clearly separate the three tiers: High-tier customers spend roughly twice as much per transaction, have roughly double the purchase history, and are far more likely to be subscribed than Low-tier customers.

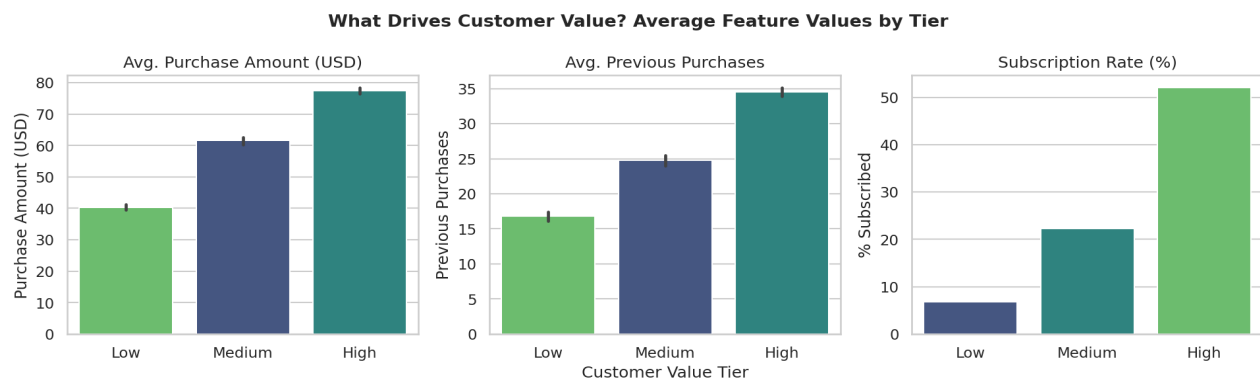


Figure 1. Average purchase amount, previous purchases, and subscription rate by customer value tier. All three increase sharply from Low to High.

Finding 2 — Subscription status is the single strongest signal. Customers without a subscription are mostly Low/Medium value (none reach High in this dataset), while subscribed customers are predominantly High value. This single yes/no field is one of the most actionable levers for the marketing team.

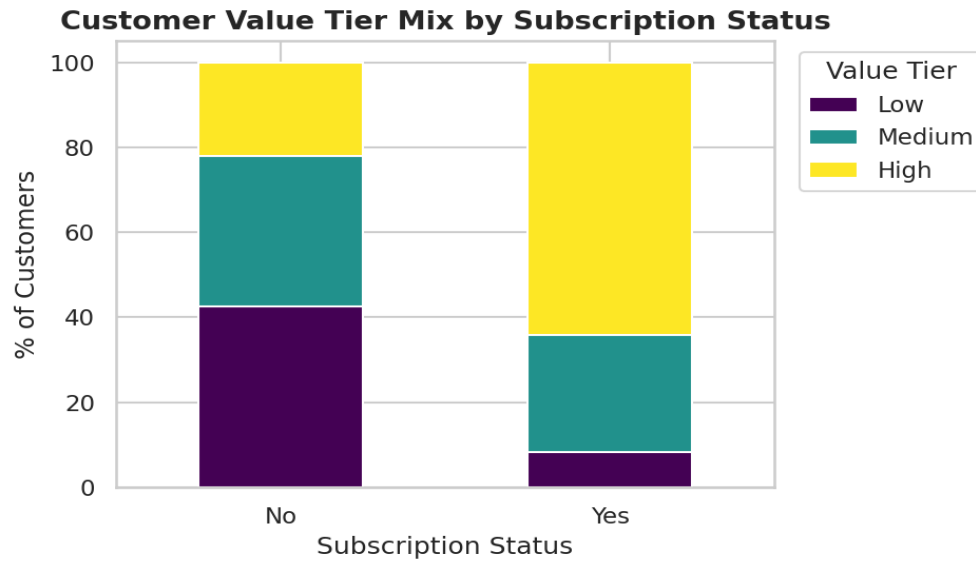


Figure 2. Customer value tier composition for subscribers vs. non-subscribers. Subscribers are overwhelmingly High value.

Finding 3 — Season, category, gender, age, and review rating are weak differentiators. These factors showed only minor variation across value tiers in the EDA, suggesting they are secondary considerations rather than primary drivers of marketing targeting decisions.

4. Model Performance

All three models were trained and evaluated on the same held-out test set (780 customers). The table below summarizes results; full details are in the accompanying model_metrics.txt / model_metrics.csv files.

Model	Accuracy	Macro F1	MAE (tier codes)
Logistic Regression	0.979	0.979	0.029
Decision Tree (depth 5)	0.897	0.898	0.153
Random Forest (100 trees)	0.924	0.925	0.117

Table 1. Model comparison on the test set. Logistic Regression was selected as the **final model** based on the highest Macro F1-score.

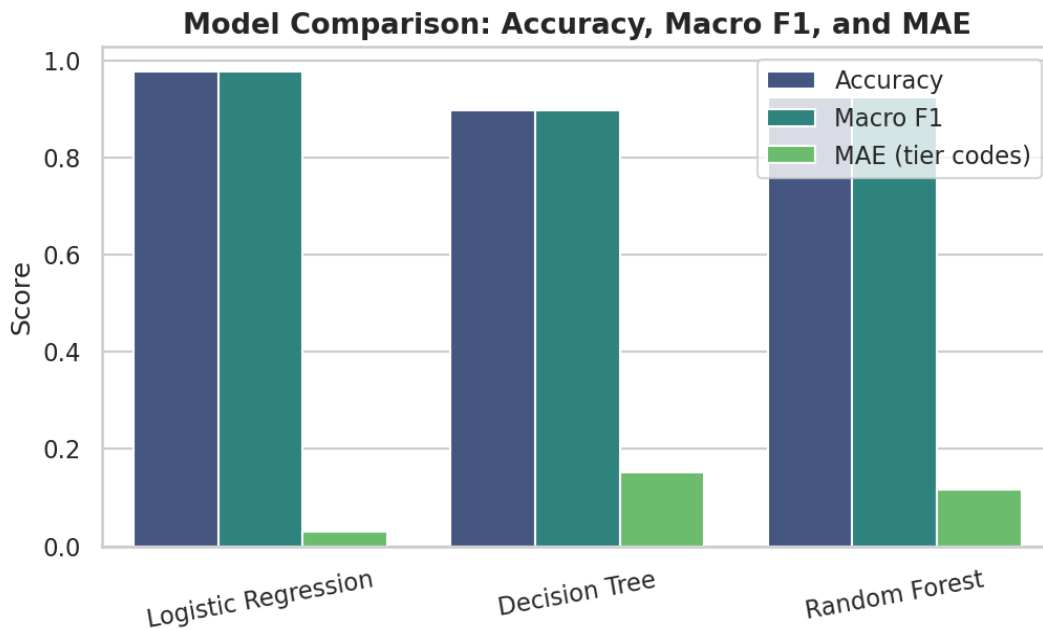


Figure 3. Side-by-side comparison of Accuracy, Macro F1, and MAE for all three models. Logistic Regression performs best on all three metrics.

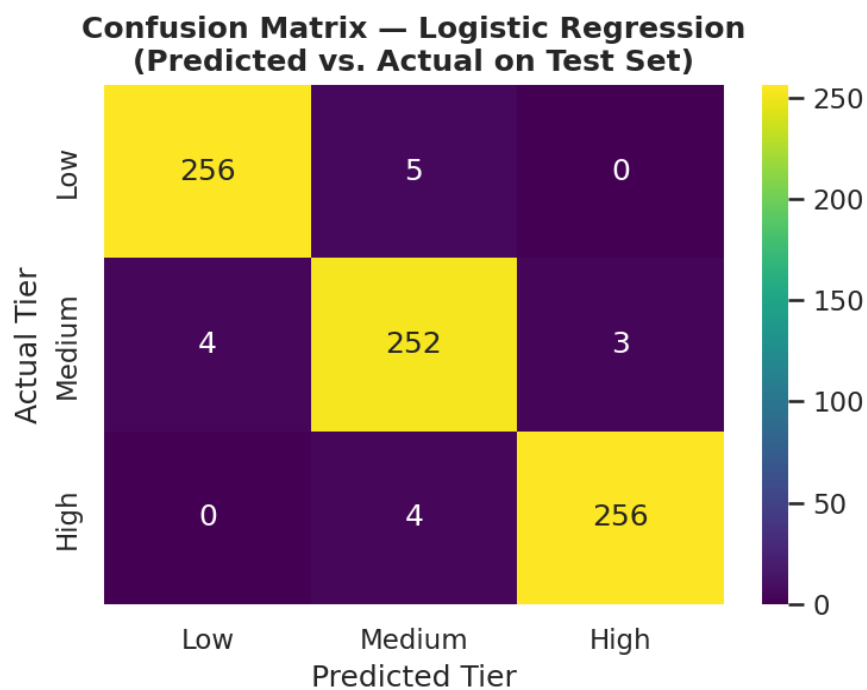


Figure 4. Confusion matrix for the final model (Logistic Regression) on the test set. The vast majority of predictions fall on the diagonal (correct), and the few misclassifications are all to an *adjacent* tier (e.g. Medium predicted as High), never a two-tier miss.

Why is performance so high? The target variable (customer_value_tier) was itself constructed from Purchase Amount, Previous Purchases, and Subscription Status — the same features available to the models. The near-perfect scores therefore confirm (rather than inflate) Finding 1: these three features are, by construction and in practice, the dominant drivers of customer value. In a live deployment, the model's

value comes from instantly scoring *new* customers into a tier the moment their spend/history/subscription data is known, without manual review.

Top 5 most influential features (final model): Subscription Status (subscribed), Purchase Amount, Previous Purchases, Frequency of Purchases (every 3 months), and Promo Code Used. This matches the EDA findings closely.

5. Recommendations

1. Prioritize subscription conversion for Medium-value, non-subscribed customers. Subscription status is the single strongest lever for moving a customer into the High-value tier. The marketing team should identify Medium-tier customers who are not subscribed and target them with subscription incentives (e.g. a free trial, loyalty discount), as this segment has the shortest path to becoming High-value.

2. Build a tiered retention program around purchase history and spend. High-value customers have substantially higher average spend and purchase history. Rather than uniform spend across all customers, allocate a disproportionate share of the retention budget (loyalty perks, early access, personalized offers) to High-tier customers to protect this segment, while using lighter-touch, lower-cost campaigns (e.g. promo codes, which showed up as a secondary signal) for Low/Medium customers to nudge incremental spend.

3. Deploy the Logistic Regression model to score the full customer base and automate the action mapping. Use the final model to assign every existing and new customer a value tier, then connect each tier to a specific action: *High* → *retention offer*, *Medium* → *conversion nudge (subscription / promo)*, *Low* → *leave alone / minimal-cost engagement only*. Because the model relies on data that's already collected at the point of sale (spend, purchase history, subscription status), this can run automatically with no additional data collection.

6. Ideas for Further Research

- Acquire true CLV or revenue-over-time data (e.g. from CRM/transaction history) to replace the proxy score with an actual financial outcome, and re-validate that the same features remain predictive.
- Test whether the 'leave alone' Low-tier segment responds positively to small, low-cost nudges (e.g. promo codes) via a controlled A/B test, rather than assuming no action is best.
- Incorporate richer seasonal/category interaction features (e.g. category preference combined with season) now that individual season and category were found to be weak on their own — interactions may reveal more nuanced segments.
- Once the model is deployed, monitor for concept drift: re-train periodically as customer behaviour and the underlying tier definitions evolve, and track whether marketing actions driven by the model actually shift customers toward higher tiers over time.

7. Conclusion

This project demonstrates a complete, reproducible pipeline — from raw data through cleaning, exploratory analysis, preprocessing, and modeling — that classifies BigMart customers into High/Medium/Low value tiers with very high accuracy (97.9% Macro F1 for the final Logistic Regression

model). The analysis consistently points to three actionable drivers of customer value: purchase amount, purchase history, and subscription status. Acting on these — especially converting Medium-value non-subscribers — offers a clear, low-friction path to the stated goal of improving marketing ROI through personalized investment strategy.